

Probabilistic Linear Solvers: Algorithmic Scaffold

► 19.1 Gaussian Distributions over Matrices

To build a concrete solver, we need a concrete family of probability distributions over matrices. And since matrices are structured objects, constructing such a family is not entirely straightforward. There are arbitrarily many ways to define a probability measure over the elements of a real matrix. To arrive at tractable forms, though, we require a probability measure that “fits well” with observations of the form $Y = AS$.

For example, the widely known Wishart distribution¹ might at first seem like a great choice to model least-squares problems (Eq. (16.1)), since it assigns probability density to only symmetric positive definite matrices.² Unfortunately, the posterior arising from conditioning on the linear projections (Y, S) is not Wishart, and has no obvious compact form. However, see Exercise 19.4 and Section 15.2 for a further discussion of how the Wishart distribution can be connected, at least in approximation, to the following derivations.

Instead of the Wishart, we will once again resort to Gaussian distributions. It is straightforward to define a joint Gaussian measure over the elements of any real matrix, even a rectangular $X \in \mathbb{R}^{N \times K}$ by using the vectorisation operation (§15.1) and using the multivariate Gaussian distribution over vectors to define³

$$\mathcal{N}(X; X_0, \Sigma_0) := \frac{\exp\left(-\frac{1}{2}(\vec{X} - \vec{X}_0)^\top \Sigma_0^{-1}(\vec{X} - \vec{X}_0)\right)}{(2\pi)^{NK/2} |\Sigma_0|^{1/2}}, \quad (19.3)$$

where $\vec{X}_0$ is any vector in $\mathbb{R}^{NK}$, interpreted as a vectorised

¹ Wishart (1928)

² The Wishart distribution $\mathcal{W}(X; V, \nu)$ is the measure over symmetric matrices $X \in \mathbb{R}^{N \times N}$ with $\nu > N - 1 \in \mathbb{R}$ *degrees of freedom* and the positive definite *scale matrix* $V \in \mathbb{R}^{N \times N}$, defined by probability density function (suppressing a complicated normalisation constant)

$$\mathcal{W}(X; V, \nu) \propto |X|^{(\nu-N-1)/2} e^{-1/2 \text{tr}(V^{-1}X)}. \quad (19.1)$$

It can also be characterised as the measure arising from the following generative process. Consider ν vectors $w_i \in \mathbb{R}^N$ drawn independently from $p(w_i) = \mathcal{N}(0, V)$. Then the symmetric $N \times N$ matrix

$$X = \sum_{i=1}^{\nu} w_i w_i^\top \quad (19.2)$$

has distribution $\mathcal{W}(X; V, \nu)$.

³ We will omit the arrow over matrices in the shorthand $\mathcal{N}(\cdot; \cdot, \cdot)$. For matrices, the Gaussian distribution will always be used in this sense, so there is no risk of ambiguity.

matrix, and Σ_0 is a symmetric, positive semi-definite matrix in $\mathbb{R}^{NK \times NK}$. If Σ_0 is full rank (i.e. if it is SPD), this distribution assigns non-zero measure to *every* matrix in $\mathbb{R}^{NK}$. In particular, if X is square, this distribution assigns non-vanishing measure to symmetric and asymmetric, to positive definite and indefinite matrices, and to invertible and non-invertible ones. Sections 19.5 and 19.6 will consider what kind of restrictions to this generality can, and cannot, be readily incorporated in a Gaussian prior. For the moment, though, we consider the general form of Eq. (19.3).

► 19.2 A Prior over A , H , or x ?

When considering probabilistic formulations of the linear problem, we have to consider what, exactly, are the uncertain aspects of the equation $Ax = b$. Is it just the vector x ? Are we looking for one solution x for one particular b , or for the general inverse $H = A^{-1}$ that can solve for any b ? In fact, we may even be uncertain about the elements of A themselves, because being able to perform multiplications $Ad = z$ for arbitrary $d \in \mathbb{R}^N$ (the only operation on A required for Algorithm 17.2) is not the same as having access to every element of A .⁴ For the majority of this chapter, we will consider the more general problem of inferring the matrix-valued objects. Section 20.2 considers simplifications arising if one only requires one particular solution x to one particular b . This means there are principally two different ways to use a Gaussian prior in the solution of a linear system. Both have historical⁵ and practical relevance.

1. One may treat the matrix A itself as the latent object, and define a joint probability distribution⁶ $p(A, Y, S)$. This model class will be called *inference on A* . This approach has the advantage that the computation of the matrix-matrix product $AS = Y$ is described explicitly. This would be relevant, for example, if the main source of uncertainty is in this computation itself – if we do not actually compute exact matrix-matrix multiplications, but only approximations of it (a setting not further discussed here).

The downside of this formulation is that it does not explicitly involve x . This is an issue because a tractable probability distribution on A may induce a complicated distribution on x . For intuition, Figure 19.1 shows distributions of the inverse of a scalar Gaussian variable of varying mean. For matrices, the situation is even more complicated, as the probability measure might put non-vanishing density on matrices that

⁴ Instead, A may be an implicitly defined linear operation. For example, a convolution operation can be defined as “take the Fourier transform of x , multiply the resulting vector element-wise, then take the Fourier back-transform”, and can be implemented without actually writing it out as an explicit matrix.

⁵ Dennis and Moré (1977), §7.3

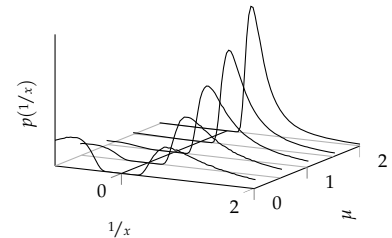


Figure 19.1: Inverses of Gaussian variables are not themselves normal distributed. The plot shows the distribution of x^{-1} if $p(x) = \mathcal{N}(x; \mu, 1)$, for five different values of μ . Since $(x + \epsilon)^{-1} \approx x^{-1} - \epsilon x^{-2}$, in the limit $|\mu| \rightarrow \infty$, the distribution approaches

$$p(x^{-1}) \approx \mathcal{N}(x^{-1}; \mu^{-1}, \mu^{-4}).$$

However, for small values of μ , the distribution becomes strongly bi-modal. It is therefore clear that we will have to resort to approximations if we want to infer both matrices and their inverse while using a Gaussian distribution to model either variable. See also Figure 19.9 for more discussion.

⁶ We will generally assume that b itself is known with certainty, and thus not explicitly include it in the generative model.

do not even have an inverse to begin with.

2. The alternative is to write the solution x explicitly as

$$x = Hb$$

(recall that we assumed A to be SPD, so the inverse $H = A^{-1}$ exists), and to formulate a distribution $p(H, Y, S)$ on H . This approach will be called *inference on H* . The advantage of this formulation is that, since x is a linear function of H , simple tractable posteriors on H translate into tractable posteriors on x . The downside is that if $Y = AS$ does not hold exactly – if the computations are performed approximately – it can be difficult to capture the likelihood $p(Y, S | H)$.

We will focus on the special but important case that observations $Z = AD$ are made without noise. In this setting, the same set of observations can be written as linear maps of either A or H :

$$Z = AD \quad \Longleftrightarrow \quad D = HZ.$$

In this case, inference on A can be directly transformed into inference on H by simultaneously exchanging

$$S \leftrightarrow Y \quad \text{and} \quad A \leftrightarrow H.$$

Because of this connection, to avoid cluttering notation, we will consider only inference on A for the moment, and return to study inference on H later.

► 19.3 General Gaussian Inference

How far can we get with a general Gaussian prior

$$p(A; A_0, \Sigma_0) = \mathcal{N}(A; A_0, \Sigma_0)$$

over the elements of A ? The observations $Y = AS$ can be cast as a Dirac likelihood, the limit of a Gaussian likelihood:

$$p(S, Y | A) = \delta(\vec{Y} - (I \otimes S^\top) \vec{A}) = \lim_{\beta \rightarrow 0} \mathcal{N}(Y; (I \otimes S^\top) \vec{A}, \beta \Lambda),$$

for any constant, SPD $\Lambda \in \mathbb{R}^{NM \times NM}$ (recall that M is the number of observations: $S \in \mathbb{R}^{N \times M}$). The Kronecker product formulation shows that the observation Y is a *linear* projection of the elements of A , hence the posterior measure arising from this conditioning observation will also have Gaussian form. From Eqs. (3.10) and (3.8), it is

$$p(A | Y, S) = \mathcal{N}(A; A_M, \Sigma_M), \quad (19.4)$$

with mean and covariance given by, respectively,

$$\vec{A}_M := \vec{A}_0 + \Sigma_0(I \otimes S) \underbrace{((I \otimes S^\top)\Sigma_0(I \otimes S))}_{=:G_M}^{-1}(\vec{Y} - (I \otimes S^\top)\vec{A}_0), \text{ and} \quad (19.5)$$

$$\Sigma_M := \Sigma_0 - \Sigma_0(I \otimes S) \underbrace{((I \otimes S^\top)\Sigma_0(I \otimes S))}_{=:G_M}^{-1}(I \otimes S^\top)\Sigma_0. \quad (19.6)$$

Unfortunately, the Gram matrix G_M featuring in both mean and covariance is of size $NM \times NM$, and hence actually *larger* than the original $N \times N$ matrix. But there is little point in an inference model for the inverse of a matrix which requires computing the inverse of an even larger matrix! Thus, some structure will have to be imposed on Σ_0 to simplify the posterior's parameters, and build a practical algorithm.

We briefly reassure ourselves that, no matter the prior covariance Σ_0 , the posterior of Eq. (19.4) is consistent with the observations, in the sense that it only puts non-zero mass on matrices $\tilde{A}$ obeying $\tilde{A}S = Y$. In this regard, we use the general property of Gaussian measures from Eq. (3.4) to see that the marginal distribution over the linear projection $\vec{AS} = (I \otimes S)^\top \vec{A}$ is

$$\begin{aligned} p(AS) &= \mathcal{N}(\vec{AS}; (I \otimes S)^\top \vec{A}_M, (I \otimes S)^\top \Sigma_M (I \otimes S)) \quad (19.7) \\ &= \mathcal{N}(\vec{AS}; Y, 0). \end{aligned}$$

Hence, any sample $\tilde{A}$ from this measure is consistent with the observations. If this sample is invertible, it also has the property $\tilde{A}^{-1}Y = S$ required in Algorithm 17.2. This holds, in particular, also for the posterior mean A_M . To arrive at a practical algorithm, we will now consider two classes of covariances. Each allows efficient inference over matrix elements, and also ensures that A_M is invertible, with an analytically computable inverse. As such, they provide a good candidate for the estimators H_i used in Algorithm 17.2.

► 19.4 Kronecker Covariances for Efficient Computation

The general Gaussian prior (19.3) is quite a flexible model; it allows for arbitrary covariance between any pair of matrix elements. To reduce computational cost, we can try to restrict the expressive power of Σ_0 .

This situation is not unrelated to that in integration. To construct tractable integration methods in Chapter II, we had to restrict the prior to GP models with mean and covariance functions on the integrand chosen such that the posterior mean could be analytically integrated. In this chapter, we will similarly be forced to restrict the prior covariance on matrix elements

to a class that allows analytic inversion of the posterior mean. While it is a somewhat ill-posed task to identify the set of all analytically integrable kernels for quadrature, we will below identify a concrete class of tractable priors for linear algebra.

A naïve choice would be to model all matrix elements as independent of each other. This corresponds to a diagonal Σ_0 , and means computing G costs $\mathcal{O}(NM^2)$. However, this leads to a G whose inversion costs more than M^3 (Exercise 19.1), which is still an unacceptably high cost.

A better prior would encode the fact that $\vec{A}$ is not just a long vector, but contains the elements of a square matrix. The projections terms $(I \otimes S)$, with their Kronecker product structure, already contain information about the generative process of the observations. We thus consider a Kronecker product for the prior covariance, too:⁷

$$\Sigma_0 = V_0 \otimes W_0 \quad \text{with SPD } V_0, W_0 \in \mathbb{R}^{N \times N}. \quad (19.8)$$

What kind of prior assumptions are we making here? If both matrices in a Kronecker product are SPD, so is their Kronecker product (see Eq. (15.6)). Hence, Eq. (19.8) yields an SPD overall covariance, and the prior assigns non-vanishing probability density to every matrix A , including non-invertible, indefinite ones, etc., despite the fact that such SPD matrices V_0, W_0 only offer

$$2 \cdot (1/2N(N+1)) = N(N+1)$$

degrees of freedom (as opposed to the $1/2N^2(N^2+1)$ degrees of freedom in a general SPD Σ_0).⁸

The prior assumptions encoded by a Kronecker product in the covariance are subtle. A few intuitive observations follow. The Kronecker covariance can be written as

$$\text{cov}(A_{ij}, A_{kl}) = [V_0 \otimes W_0]_{ij,kl} = [V_0]_{ik} \cdot [W_0]_{jl}. \quad (19.9)$$

As such, it is tempting to think of V_0 as capturing covariance “among the rows” and W_0 “among the columns” (Figure 19.2). This is not entirely incorrect. Consider the following process to create a matrix $\vec{A}$:

1. Draw N column-vectors $b_i \in \mathbb{R}^{N \times 1}, i = 1, \dots, N$ i.i.d. as

$$b_i \sim \mathcal{N}(0, V_0).$$

2. Draw N row-vectors $c_j \in \mathbb{R}^{1 \times N}, i = 1, \dots, N$ i.i.d. as

$$c_j \sim \mathcal{N}(0, W_0).$$

Exercise 19.1 (intermediate, and somewhat tedious. Solution on p. 362). Consider the assumption $\Sigma_{ij,kl} = \delta_{ik}\delta_{jl}W_{ij}$, with a positive definite W . It amounts to an independent Gaussian prior, of variance W_{ij} , for every single matrix element. Show that this choice gives rise to a G in Eq. (19.5) whose inverse involves N separate positive definite matrices, each of size $M \times M$.

⁷ Distributions of the form $\mathcal{N}(X; X_0, V \otimes W)$ are sometimes called a *matrix-variate normal*, due to a paper by Dawid (1981). This convention will be avoided here, since it can give the incorrect impression that this is the only possibility to assign a Gaussian distribution over the elements of a matrix, when in fact Eq. (19.3) is the most general such distribution.

⁸ One helpful intuition for this situation is to convince oneself that the space of Kronecker products spans a sub-space of rank one within the space of $N^2 \times N^2$ real matrices, and that this sub-space does contain a space of SPD matrices.

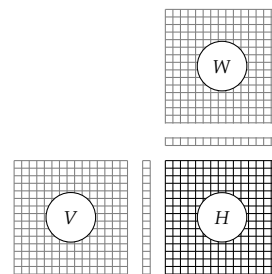


Figure 19.2: Imposing Kronecker product structure $V \otimes W$ on the prior covariance means the covariance over matrix elements is governed by two factors. V affects covariance as a function of row number, W as a function of column number. This is not quite the same as assuming the matrix itself is a product of two terms.

3. Set $\tilde{A}_{ij} = b_i c_j + A_{0,ij}$ i.e. $\tilde{A} = B \odot C + A_0$, if B and C denote the matrices resulting from arranging the vectors b_i and c_j into square matrices in the obvious way.

The matrix $\tilde{A}$ arising from this process is *not* Gaussian distributed.⁹ But it indeed¹⁰ satisfies Eq. (19.9): the covariance is Kronecker-structured.

Another helpful observation is that the marginal variance of individual matrix elements under the choice $\Sigma_0 = V_0 \otimes W_0$ is determined solely by the diagonals of the two matrices in the Kronecker product:

$$\text{var}(A_{ij}) = [V_{A0}]_{ii} \cdot [W_{A0}]_{jj}.$$

Figure 19.3 shows five samples each from two different Gaussian distributions with Kronecker product covariance.

The main takeaway is that while the Kronecker covariance does represent a helpful restriction, it is not one that limits the space of matrices that we can infer. The prior measure encompasses all matrices.

What Kronecker structure on the covariance *does* achieve is a drastic reduction of computational complexity. Choosing Σ_0 as in Eq. (19.8) shortens the posterior's mean and covariance (Eqs. (19.5) and (19.6)) to, respectively,

$$A_M = A_0 + \underbrace{(Y - A_0 S)}_{=: \Delta_M \in \mathbb{R}^{N \times M}} \underbrace{(S^T W_0 S)^{-1}}_{\in \mathbb{R}^{M \times M}} \underbrace{S^T W_0}_{\in \mathbb{R}^{M \times N}}, \quad \text{and (19.10)}$$

$$\Sigma_M = V_0 \otimes \underbrace{(W_0 - W_0 S (S^T W_0 S)^{-1} S^T W_0)}_{=: W_M}. \quad (19.11)$$

Note the absence of arrows over A_M and H_M . The posterior mean A_M is a sum of the prior mean (which can, of course, be chosen to be simple, e.g. a scalar or diagonal matrix), and an *outer product* of rank at most M . All the objects can be stored in $\mathcal{O}(NM^2)$, and they contain a single matrix inverse of only an $M \times M$ matrix. The terms in this outer product are:

- Δ_M the residual between the expected value $A_0 S$ (the prediction for Y under the prior), and the actual observation,
- $S^T W_0 S$ the predictive covariance between the rows of the residual, and
- $S^T W_0$ the predictive covariance between the residual's rows and the rows of A .

⁹ NB: the product of two Gaussian probability distributions is another Gaussian distribution (times a Gaussian normalisation constant). But the product of two Gaussian *random variables* is *not* itself a Gaussian random variable!

¹⁰ This is because

$$\begin{aligned} \mathbb{E}(\tilde{A}_{ij} \tilde{A}_{k\ell}) - \mathbb{E}(\tilde{A}_{ij}) \mathbb{E}(\tilde{A}_{k\ell}) \\ = \mathbb{E}(a_i b_j a_k b_\ell) = [V_0]_{ik} \cdot [W_0]_{j\ell}. \end{aligned}$$

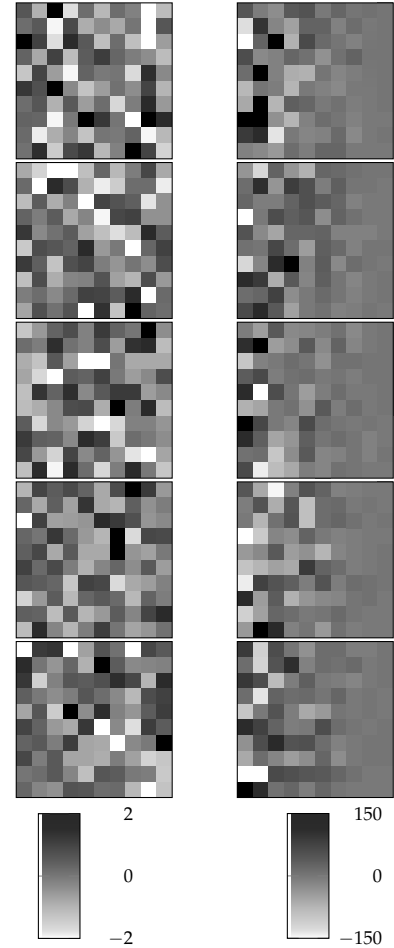


Figure 19.3: Five i.i.d. samples from the distribution $\mathcal{N}(I, V_0 \otimes W_0)$ for the choices $V_0 = W_0 = I$ (left column) and $V_0 = \text{diag}[10^4, 9^4, 8^4, \dots]$, $W_0 = I$ (right column), respectively.

Exercise 19.2 (easy). Using Eqs. (15.2) to (15.7), prove that the choice (19.8) simplifies Eqs. (19.5) & (19.6) to (19.10) & (19.11), respectively.

The vector Y shows up in A_M without a mapping to its left. In the simple case of $A_0 = 0$, the columns of A_M are a scaled copy of Y , with each rows transformed by W_0 . This is reflected in the posterior variance: Observing linear projections of all the rows of A , given assumptions about the covariance along rows and columns, affects the “uncertainty along the rows”, but not “along the columns”.

The posterior mean A_M of Equation (19.10) provides an estimator for A . Algorithm 17.2 instead requires an estimator for $H = A^{-1}$. Thanks to the low-rank structure of A_M , however, it is straightforward to compute its inverse, and use it as such an estimator $H_i = A_i^{-1}$. Using the matrix inversion lemma (15.9), we find, after a few simple re-arrangements,

$$\begin{aligned} A_M^{-1} &= (A_0 + \Delta_{AM}(S^T W_0 S)^{-1} S^T W_0)^{-1} \\ &= A_0^{-1} + (S - A_0^{-1} Y)(S^T W_0 A_0^{-1} Y)^{-1} S^T W_0 A_0^{-1}. \end{aligned} \quad (19.12)$$

But we remind ourselves that this inverse of the expected value of A is not the same as the expected value of the inverse! The latter is more challenging to compute. Nevertheless the following result shows that, for sensible choices of the prior parameters A_0 and W_0 , the estimate A_M^{-1} exists.

Lemma 19.3. *Assume A_0 and W_0 are SPD, and the search directions S are chosen to be linearly independent. Then, for our assumption of SPD A , the inverse (19.12) exists.*

Proof. If A_0 is SPD, its inverse exists. $Y = AS$, and products of SPD matrices are SPD. Thus, $W_0 A_0^{-1} A$ is SPD, hence $S^T W_0 A_0^{-1} AS$ invertible. $\square$

We thus have our first way to concretely realise Algorithm 17.2: Its line 12 can be implemented by assigning the Gaussian prior

$$p(A) = \mathcal{N}(A; A_0, V_0 \otimes W_0), \quad \text{with } A_0, V_0, W_0 \text{ SPD}, \quad (19.13)$$

to A , which gives rise to the posterior mean of Eq. (19.10). If the directions S are independent, the inverse of the mean is given by Eq. (19.12) and provides the estimator H_i .¹¹ That is, explicitly,

$$\begin{aligned} H_i &= (A_0 + \Delta_{AM}(S^T W_0 S)^{-1} S^T W_0)^{-1} \\ &= A_0^{-1} + (S - A_0^{-1} Y)(S^T W_0 A_0^{-1} Y)^{-1} S^T W_0 A_0^{-1}. \end{aligned}$$

¹¹ If the columns of S are *not* linearly independent, this could be rectified internally in the solver by keeping track of a linearly independent sub-space. Doing so would require extra linear algebra operations (for example it could be realised using the SVD of S).

► 19.5 Encoding Symmetry

We defined the problem $Ax = b$ with the assumption that A is a symmetric positive definite matrix. The priors on A or A^{-1} considered so far, however, do not encode this knowledge yet. They

assign non-zero probability density to all real-valued matrices. This is not just an aesthetic shortcoming: §18.2 guarantees conjugacy of the search directions if the estimator H_M is symmetric; but the estimator of Eq. (19.12) is not symmetric in general.

We will now find that *symmetry* can be encoded in the prior, with a bit of legwork, without deviating from tractable Gaussian inference. This is because symmetry is a *linear* constraint; it can be written as the set of linear equations

$$A_{ij} - A_{ji} = 0, \quad \forall 1 \leq i, j \leq N.$$

From Eq. (3.4), we recall that the Gaussian family is closed under such constraints. In contrast, positive definiteness is a considerably more intricate property because the positive definite cone is not a linear sub-space of $\mathbb{R}^{N^2}$. A discussion of positive definiteness follows in §19.6.

To capture symmetry, we will begin by simplifying the prior covariance from Eq. (19.13) to the choice $V_0 = W_0$, i.e.

$$p(A) = \mathcal{N}(A; A_0, W_0 \otimes W_0). \quad (19.14)$$

This step alone does not encode symmetry (samples from this distribution are still asymmetric with probability one), but it avoids technical complications in the following.

For a formal treatment, we introduce two projection operators acting on the space $\mathbb{R}^{N^2}$ of square $N \times N$ matrices:

$\Pi_{\ominus} : \mathbb{R}^{N^2} \rightarrow \mathbb{R}^{N^2}$, with elements

$$\Pi_{\ominus} (ij), (k\ell) := 1/2(\delta_{ik}\delta_{j\ell} + \delta_{i\ell}\delta_{jk}),$$

is the *projection onto the space of symmetric matrices*. It has the characteristic property

$$\natural \Pi_{\ominus} \vec{X} = 1/2(X + X^{\top}).$$

$\Pi_{\oplus} : \mathbb{R}^{N^2} \rightarrow \mathbb{R}^{N^2}$, with elements

$$\Pi_{\oplus} (ij), (k\ell) := 1/2(\delta_{ik}\delta_{j\ell} - \delta_{i\ell}\delta_{jk}),$$

is the *projection onto the space of anti-symmetric¹² matrices*. It has the characteristic property

$$\natural \Pi_{\oplus} \vec{X} = 1/2(X - X^{\top}).$$

It is easy to convince oneself that $\Pi_{\ominus}$ and $\Pi_{\oplus}$ are orthogonal projection operators that jointly span $\mathbb{R}^{N^2}$, i.e. that¹³

¹² Also known as *skew-symmetric* matrices.

¹³ This also implies that every matrix $X \in \mathbb{R}^{N \times N}$ can be written as a sum of a symmetric and an antisymmetric part:

$$\vec{X} = \Pi_{\ominus} \vec{X} + \Pi_{\oplus} \vec{X},$$

which holds simply because

$$X = 1/2(X + X^{\top} + X - X^{\top}).$$

$$\begin{aligned}
\Pi_{\ominus}\Pi_{\ominus} &= \Pi_{\ominus}, & \Pi_{\oplus}\Pi_{\oplus} &= \Pi_{\oplus}, & \text{and} \\
\Pi_{\ominus}^{\top} &= \Pi_{\ominus}, & \Pi_{\oplus}^{\top} &= \Pi_{\oplus}, & \text{and} \\
\Pi_{\ominus}\Pi_{\oplus} &= \Pi_{\oplus}\Pi_{\ominus} = 0_{N^2}, & \Pi_{\ominus} + \Pi_{\oplus} &= I_{N^2}. & (19.15)
\end{aligned}$$

We also notice that, for Kronecker products of a matrix W with itself, we have (leaving out a few simple lines of algebra)

$$\begin{aligned}
[\Pi_{\ominus}(W \otimes W)\Pi_{\oplus}]_{(ij),(k\ell)} \\
= \frac{1}{4}(W_{ik}W_{j\ell} - W_{i\ell}W_{jk} - W_{ik}W_{j\ell} + W_{i\ell}W_{jk}) = 0.
\end{aligned}$$

So a Kronecker product of this type can be written as a direct sum of a “symmetric” part and an “anti-symmetric” part:

$$W \otimes W = \underbrace{\Pi_{\ominus}(W \otimes W)\Pi_{\ominus}}_{=:W \otimes W} + \underbrace{\Pi_{\oplus}(W \otimes W)\Pi_{\oplus}}_{=:W \otimes W}.$$

These two products are known¹⁴ as the *symmetric Kronecker product* $W \otimes W$ and the *skew-symmetric Kronecker product* $W \otimes W$, with elements

$$\begin{aligned}
[C \otimes D]_{(ij),(k\ell)} &= 1/2(C_{ik}D_{j\ell} + C_{i\ell}D_{jk}), \text{ and} & (19.16) \\
[C \otimes D]_{(ij),(k\ell)} &= 1/2(C_{ik}D_{j\ell} - C_{i\ell}D_{jk}).
\end{aligned}$$

They inherit some, but not all, of the great properties of the Kronecker product. In particular, when applied to symmetric matrices,¹⁵

$$(W \otimes W)^{-1} = W^{-1} \otimes W^{-1}.$$

However,¹⁶ for general C and D ,

$$(C \otimes D)^{-1} \neq C^{-1} \otimes D^{-1}.$$

We also have

$$\begin{aligned}
(C \otimes D)\vec{X} &= 1/2(CXD^{\top} + CX^{\top}D^{\top}), \text{ and} \\
(C \otimes D)\vec{X} &= 1/2(CXD^{\top} - CX^{\top}D^{\top}). & (19.17)
\end{aligned}$$

Using this framework, the *information* about A 's symmetry can be explicitly written as an observation with likelihood

$$p(\Theta | A) = \delta(\Pi_{\oplus}\vec{A} - 0) = \lim_{\beta \rightarrow 0} \mathcal{N}(\vec{0}_{N^2}; \Pi_{\oplus}\vec{A}, \beta I_{N^2}). \quad (19.18)$$

By Eqs. (3.8) and (3.10), the density on A resulting from conditioning the prior (19.14) with this likelihood is

$$\begin{aligned}
p(A | \Theta) &= \mathcal{N}(A; \vec{A}_0 - \Sigma_0 \Pi_{\oplus}^{\top} (\Pi_{\oplus} \Sigma_0 \Pi_{\oplus}^{\top})^{-1} (-\Pi_{\oplus} \vec{A}_0), \\
&\quad \Sigma_0 - \Sigma_0 \Pi_{\oplus}^{\top} (\Pi_{\oplus} \Sigma_0 \Pi_{\oplus}^{\top})^{-1} \Pi_{\oplus} \Sigma_0). & (19.19)
\end{aligned}$$

¹⁴ van Loan (2000)

Exercise 19.4 (easy). Show that for matrices X drawn from the Wishart distribution $\mathcal{W}(X; V, \nu)$ (defined in Eq. (19.1)) the elements of X have the covariance

$$\text{cov}(X_{ij}, X_{k\ell}) = 2\nu(V \otimes V).$$

Hint: Use the generative definition of Eq. (19.2). The fourth moment of a central normal distribution is given by Isserlis' theorem (Isserlis, 1918):

$$p(a) = \mathcal{N}(a; 0, \Sigma) \Rightarrow$$

$$\mathbb{E}(a_i a_j a_k a_{\ell}) = \Sigma_{ij} \Sigma_{k\ell} + \Sigma_{ik} \Sigma_{j\ell} + \Sigma_{i\ell} \Sigma_{jk}.$$

¹⁵ If $W \in \mathbb{R}^{N \times N}$ is of full rank, the matrix $W \otimes W$ has rank $1/2 N(N+1)$, the dimension of the space of all real symmetric $N \times N$ matrices. That its inverse on that space is given by $W^{-1} \otimes W^{-1}$ can be seen from Eq. (19.17). The inverse on asymmetric matrices is not defined.

¹⁶ Alizadeh, Haeberley, and Overton (1988)

Using the Kronecker structure $\Sigma_0 = W_0 \otimes W_0$ and the identity from Eq. (19.15), the mean reduces to its symmetric part:

$$\begin{aligned}\vec{A}_0 - (\Pi_{\oplus} + \Pi_{\ominus})\Sigma_0\Pi_{\oplus}^T(\Pi_{\oplus}\Sigma_0\Pi_{\oplus}^T)^{-1}\Pi_{\oplus}\vec{A}_0 \\ = (\Pi_{\oplus} + \Pi_{\ominus})\vec{A}_0 - \Pi_{\oplus}\vec{A}_0 = \Pi_{\ominus}\vec{A}_0,\end{aligned}$$

while the covariance simplifies to a symmetric Kronecker product:

$$\Sigma_0 - \Sigma_0\Pi_{\oplus}^T(\Pi_{\oplus}\Sigma_0\Pi_{\oplus}^T)^{-1}\Pi_{\oplus}\Sigma_0 = \Pi_{\ominus}\Sigma_0\Pi_{\ominus} = W_0 \otimes W_0.$$

The resulting probabilistic model

$$p(A) = \mathcal{N}(A; A_0, W_0) \otimes W_0 \quad (19.20)$$

with a symmetric prior mean matrix A_0 , only assigns non-zero measure to symmetric matrices; samples from Eq. (19.20) are depicted in Figure 19.4. Like for the Kronecker product covariance, this prior, too, can be conditioned to the linear observations $Y = AS$, giving a posterior

$$\begin{aligned}p(A | Y, S, \Theta) &= \frac{\delta(Y - AS)\mathcal{N}(A; A_0, W_0 \otimes W_0)}{\int \delta(Y - AS)\mathcal{N}(A; A_0, W_0 \otimes W_0) dA} \\ &= \mathcal{N}(A; A_M, \Sigma_M).\end{aligned}$$

Some algebraic footwork¹⁷ is required to find the posterior means and covariance, the analogues to Eqs. (19.10) and (19.11). They are

$$\begin{aligned}A_M &= A_0 + (Y - A_0S)(S^TW_0S)^{-1}S^TW_0 \\ &\quad + W_0S(S^TW_0S)^{-1}(Y - A_0S)^T \\ &\quad - W_0S(S^TW_0S)^{-1}S^T(Y - A_0S)(S^TW_0S)^{-1}S^TW_0,\end{aligned} \quad (19.21)$$

$$\Sigma_M = W_M \otimes W_M, \text{ with (as in Eq. (19.11)),} \quad (19.22)$$

$$W_M := W_0 - W_0S(S^TW_0S)^{-1}S^TW_0.$$

These expressions, in particular the posterior mean A_M , play a central role not just in linear solvers, but also in nonlinear optimisation. Let us take a closer look. We first note that A_M is indeed symmetric if A_0 is symmetric, because $S^TY = S^TAS$ is symmetric. What is less obvious is that the expression added to A_M is of at most rank $2M$. This can be seen by defining the helpful terms

$$\begin{aligned}U &:= W_0S(S^TW_0S)^{-1} \in \mathbb{R}^{N \times M} \quad \text{and} \\ V &:= (I - 1/2UST)(Y - A_0S) \in \mathbb{R}^{N \times M}.\end{aligned} \quad (19.23)$$

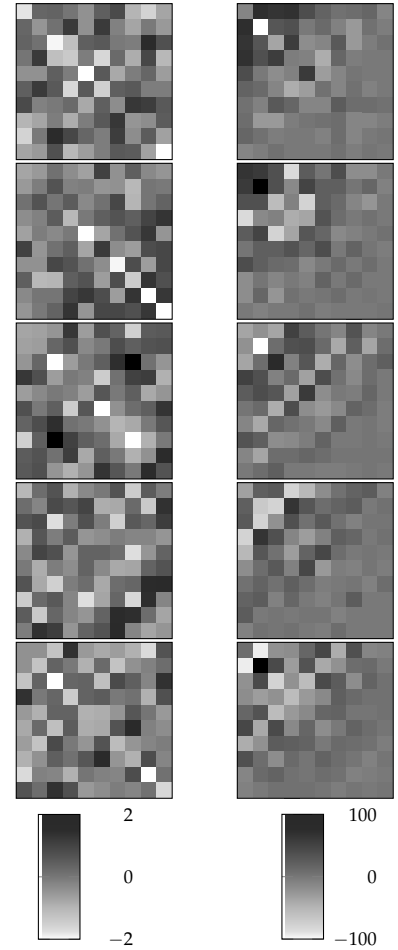


Figure 19.4: Samples from a Gaussian prior encoding symmetry. Results analogous to Figure 19.3. Five i.i.d. samples from the distribution of Eq. (19.20) for $W_0 = I$ (left column) and $W_0 = \text{diag}[10^2, 9^2, 8^2, \dots]$ (right column), respectively. Note differing choice for W_0 relative to Figure 19.3, since W_0 here is in both terms of the product.

Exercise 19.5 (moderate). Explicitly compute the evidence term

$$\int \delta(Y - AS)\mathcal{N}(A; A_0, W_0 \otimes W_0) dA.$$

What is its form?

¹⁷ For a derivation, see Hennig (2015).

Exercise 19.6 (hard). Derive the result in Eq. (19.21). In performing the derivation, try to gain an intuition for why the posterior mean (19.21) is not simply the symmetrised form $\Pi_{\ominus}A_M$ of the posterior mean from Eq. (19.10) (consider Eq. (19.19) for a hint).

Using these, A_M can be written as

$$A_M = A_0 + UV^\top + VU^\top = A_0 + \begin{bmatrix} U & V \end{bmatrix} \begin{bmatrix} \mathbf{0} & I_M \\ I_M & \mathbf{0} \end{bmatrix} \begin{bmatrix} U^\top \\ V^\top \end{bmatrix}.$$

The inverse of this expression, if it exists, follows from the matrix inversion lemma:

$$A_M^{-1} = A_0^{-1} - A_0^{-1} \begin{bmatrix} U & V \end{bmatrix} \begin{bmatrix} U^\top A_0^{-1} U & U^\top A_0^{-1} V + I \\ V^\top A_0^{-1} U + I & V^\top A_0^{-1} V \end{bmatrix}^{-1} \begin{bmatrix} U^\top \\ V^\top \end{bmatrix} A_0^{-1}. \quad (19.24)$$

There is no particularly enlightening way to simplify this expression for general model parameters A_0, W . But note that the matrix to be inverted on the right-hand side is of size $2M \times 2M$. So assuming the inverse of the prior mean A_0 is known or easy to compute, computation of this inverse estimator has complexity of *at most* $\mathcal{O}(M^3)$ (and multiplying with it has cost $\mathcal{O}(NM^2)$). Below we will find that there are special instances for which the complexity is significantly lower.

► 19.6 What about Positive Definiteness?

Encoding symmetry in the prior is analytically feasible because it amounts to a linear constraint (Eq. (19.18)). Since we assume throughout this chapter that A is not just symmetric, but also positive definite, it would be desirable to also encode this information in the prior. Unfortunately, the space of positive definite matrices is a cone, a nonlinear sub-space of $\mathbb{R}^{N^2}$, the space of all (vectorised) square $N \times N$ matrices, and also a nonlinear sub-space of $\mathbb{R}^{1/2N(N+1)}$, the space of symmetric such matrices (see Figure 19.5). Information about positive definiteness can thus not be captured in a Gaussian likelihood term using only linear terms of A .

It is, however, possible to scale the parameters of the Gaussian prior post hoc to ensure that the posterior mean estimate always lies within the positive definite cone. This is helpful in so far as it means this posterior point estimate can be trusted to be admissible, and this is how this correction is used in practice. From a probabilistic perspective, however, this is not particularly satisfying since it means the model cannot make use of the known positive definiteness during inference.

The following is a minor generalisation of a derivation in a seminal review by Dennis and Moré (1977, §7.2) reproduced in some detail here because it provides valuable insights, also used in Chapter IV. Consider the symmetry-encoding prior (19.20),

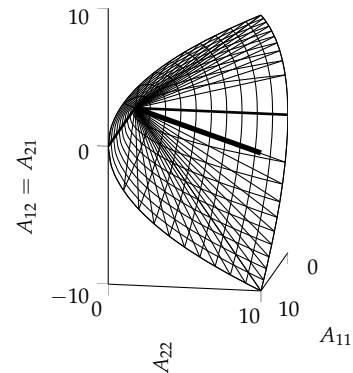


Figure 19.5: Outer boundaries of the positive definite cone within the space of symmetric 2×2 matrices (the only case that allows a plot). The thick line down the centre of the cone marks scalar matrices. The outer edge of the 2×2 positive definite cone is given by matrices A with $A_{12} = A_{21} = \pm\sqrt{A_{11}A_{22}}$.

which gives rise to the posterior mean of Eq. (19.21). Assume that the prior mean A_0 has been chosen to be symmetric positive definite. Since the posterior mean is of the same algebraic form (Gaussian, with a symmetric Kronecker product posterior covariance), the posterior can equivalently be computed iteratively, as rank-2 updates to the mean estimator. Consider the posterior belief

$$p(A \mid Y_i) = \mathcal{N}(A; A_i, W_i \otimes W_i),$$

conditioned on observations Y_i , the i matrix-vector multiplications $y_j = As_j$ for $j = 1, \dots, i$.

Using $p(A \mid Y_i)$, and given the next observation $y_{i+1} = As_{i+1}$, we can calculate a Gaussian posterior on A with the mean and covariance

$$\begin{aligned} A_{i+1} &= A_i + \frac{(y_i - A_i s_i) s_i^T W_i + W_i s_i (y_i - A_i s_i)^T}{s_i^T W_i s_i} \\ &\quad - \frac{(y_i - A_i s_i)^T s_i}{(s_i^T W_i s_i)^2} W_i s_i s_i^T W_i \\ &= A_i + uv^T + vu^T, \end{aligned} \quad (19.25)$$

for (see Eq. (19.23))

$$\begin{aligned} u &:= \frac{W_i s_i}{s_i^T W_i s_i}, \\ v &:= (y_i - A_i s_i) - \frac{W_i s_i s_i^T (y_i - A_i s_i)}{2s_i^T W_i s_i}, \\ W_{i+1} &= W_i - \frac{W_i s_i s_i^T W_i}{s_i^T W_i s_i}. \end{aligned}$$

Each posterior mean update is thus a rank-2 update. This iterative form is more manageable from an analytic perspective than the immediate form of Eq. (19.21). The idea is now to ask for a value of W_0 such that A_i can be shown by induction to be positive definite, a notion that Dennis and Moré call *hereditary positive definiteness*. For this we make use of a result from matrix perturbation theory.¹⁸ Intuitively speaking, a rank-1 update can at most shift the eigenvalues of the original matrix up or down to the value of the nearest neighbouring eigenvalues.

Lemma 19.7. *Let $A \in \mathbb{R}^{N \times N}$ be symmetric with eigenvalues*

$$\lambda_1 \leq \lambda_2 \leq \dots \leq \lambda_N.$$

And let $A^ = A + \alpha a a^T$, with $a \in \mathbb{R}^N, \alpha \in \mathbb{R}$. If $\alpha \geq 0$, then A^* has eigenvalues λ^* such that*

$$\lambda_1 \leq \lambda_1^* \leq \lambda_2 \leq \dots \leq \lambda_N \leq \lambda_N^*.$$

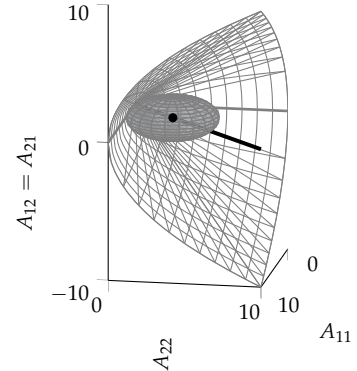


Figure 19.6: A Gaussian prior measure of mean $A_0 = 3I$ and symmetric Kronecker covariance with $W = 3I$ shown relative to the positive definite cone. The symmetric Kronecker product inherits *some* of the cone's structure in so far as the marginal variance of off-diagonal elements under this prior is half that of diagonal elements. But the distribution still assigns non-vanishing measure to the indefinite matrices outside of the cone.

¹⁸ Wilkinson (1965), pp. 95–98

If $\alpha \leq 0$, then the eigenvalues of A^* can be arranged so that

$$\lambda_1^* \leq \lambda_1 \leq \lambda_2^* \leq \cdots \leq \lambda_N^* \leq \lambda_N.$$

Now note that the rank-2 update in Eq. (19.25) can be written as the sum of two symmetric rank-1 matrices:

$$A_{i+1} = A_i + 1/2((u+v)(u+v)^\top - (u-v)(u-v)^\top). \quad (19.26)$$

This insight can be used to arrive at the following helpful statement about rank-2 updates.

Lemma 19.8 (Dennis and Moré, Thm. 7.5). *Consider the updated estimator A_{i+1} as defined in Eq. (19.25). Assuming the A_i is positive definite, and $s_i^\top W_i s_i \neq 0$, the matrix A_{i+1} is positive definite if and only if $\det A_{i+1} > 0$.*

To leverage this result, we make use of the matrix determinant lemma, Eq. (15.11). Conveniently rearranged for the present context, it reads

$$\det(I + ab^\top + ba^\top) = (1 + a^\top b)^2 - (a^\top a)(b^\top b).$$

Thus, the determinant of the updated A_{i+1} is

$$\det A_{i+1} = \det A_i \left(\frac{(y_i^\top A_i^{-1} W_i s_i)^2 - (y_i^\top A_i^{-1} y_i)(s_i^\top W_i A_i^{-1} W_i s_i) + (s_i^\top W_i A_i^{-1} W_i s_i)(y_i^\top s_i)}{(s_i^\top W_i s_i)^2} \right).$$

Since W_i is positive semi-definite, by Lemma 19.8, A_{i+1} is thus symmetric positive definite if and only if

$$y_i^\top s_i > y_i^\top A_i^{-1} y_i - \frac{(y_i^\top A_i^{-1} W_i s_i)^2}{s_i^\top W_i A_i^{-1} W_i s_i}. \quad (19.27)$$

There are two interesting special cases for which this condition can be simplified. They are summarised in the following two statements.

Corollary 19.9. *Assume A_0 is symmetric positive definite, A_i is the posterior mean estimator of Eq. (19.21), and Algorithm 17.2 is used to construct search directions $s_i, i = 1, \dots$ (in particular, this means the search directions are conjugate). If W_i has the property that $W_i s_i = y_i$, then all A_i are symmetric positive semi-definite. (This is the case for the unrealistic, but conceptually interesting parameter choice $W_0 = A$).*

Proof of Corollary 19.9. Under the assumptions of the Corollary, the right-hand side of Eq. (19.27) vanishes, and the condition is *always* fulfilled: by assumption A is SPD and $y_i^\top s_i = s_i^\top A s_i$. Algorithm 17.2 ensures the search directions s_i are conjugate under A , and thus

$$W_i s_i = A s_i - Y(S^\top Y)^{-1} Y^\top s_i = y_i - 0.$$

□

Theorem 19.10 (proof on p. 187). *Assume $A_0 = \alpha I, W_0 = \beta I$ for $\alpha, \beta \in \mathbb{R}_+$, A_i is the posterior estimator of Eq. (19.21), and the search-directions S are conjugate to each other. Then there exists a finite $\alpha_0 > 0$, so that any choice $\alpha > \alpha_0$ ensures that all A_i are positive semi-definite.*

In general, the posterior mean is not guaranteed to be positive definite (see Figure 19.7). But Corollary 19.9 and Theorem 19.10 establish two different paths to guarantee positive definiteness of the posterior mean: Corollary 19.9 establishes that if we choose W such that $W_0 S = Y$ and set any positive definite prior mean A_0 , then inference on a positive definite A will always produce positive definite posterior means (see Figure 19.8). If we consider scalar covariances instead, then Theorem 19.10 provides the weaker statement that, in that setting, it is at least possible to “drag the posterior mean into the positive definite cone” by increasing prior covariance.

Both statements, however, are dissatisfying from the probabilistic perspective for two reasons. First, they are just statements about the mean. The posterior *distribution*, being Gaussian, will always put non-zero measure on parts of the real vector-space outside of the positive definite cone. Second, these statements are of a post-hoc nature, literally: the prior still puts mass outside of the cone (see Figure 19.6). The information that A is positive definite, available *a priori*, can thus not be leveraged by the solver in its action policy. The real value of prior information is that it can change the way the algorithm acts, not just the final estimate. At the time of writing, there is no clear solution to this problem.

► 19.7 Summary: Gaussian Linear Solvers

We are interested in Gaussian inference models that can be made consistent with the evaluation strategy and the estimates constructed by iterative linear solvers for symmetric positive definite matrices. The derivations of the preceding pages show that even these comparably concrete constraints still allow for a diverse space of models. Table 19.1 summarises the four different candidates for a Gaussian framework of linear solvers. Aiming to solve $Ax = b$ for x , assuming that A is symmetric positive definite, we adopt the algorithmic paradigm of an iterative solver, as defined in Table 17.2, constructing projection-observation pairs $S = [s_1, \dots, s_M] \in \mathbb{R}^{N \times M}$ and $Y = AS = [y_1, \dots, y_M] \in \mathbb{R}^{N \times M}$.

To endow that algorithm class with a probabilistic meaning, we might directly model the matrix inverse H . Modelling H allows a joint Gaussian model over both H and the solution x . Alternatively, we might model the matrix A . Modelling A allows direct treatment of Gaussian observation noise, and still

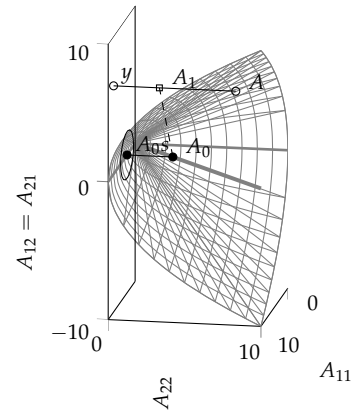


Figure 19.7: Gaussian inference under the Gaussian prior of Figure 19.6 and the projection $s = [1, 0]^T$, on the SPD matrix A with $A_{11} = A_{22} = 9, A_{12} = 0.7 \cdot 9$ (black circle). The plots on the “left wall” of the plot show the projections of the prior and A into the observation space $[A_{11}, A_{12}]^T$. Although both the prior mean and the true matrix are symmetric positive definite, the posterior mean (black square, connected to A_0 by a dashed line) lies outside of the cone. Theorem 19.10 shows that one way to fix this is to increase the prior mean. The graphical representation of this result is that A_1 always lies on the black projection line connecting A and y (recommended instant exercise: why?). As the prior mean increases, A_1 eventually moves along that line into the cone.

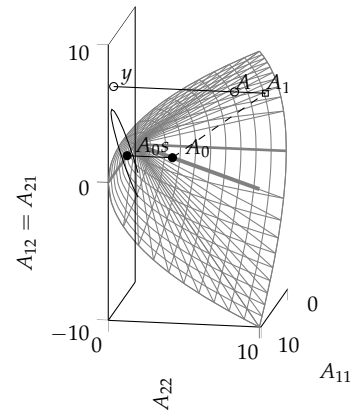


Figure 19.8: Analogous to Figure 19.7, but with the covariance choice $W = A$ considered in Corollary 19.9. Under this choice, the posterior mean always lies “to the right” of the true A along the projection line, thus in the positive definite cone.

	Asymmetric model	Symmetric model
Model for H	$p(H) = \mathcal{N}(H_0, V_0 \otimes W_0)$ $p(Y, S H) = \delta(S - HY)$ $= \lim_{\gamma \rightarrow 0} \mathcal{N}(\vec{S}; \vec{AY}, \gamma^2(I_N \otimes I_M))$ $\mathbb{E}_{ Y}(H) = H_0 + (S - H_0 Y) \underbrace{(Y^\top W_0 Y)^{-1} Y^\top W_0}_{=: U^\top}$ $\mathbb{V}_{ Y}(H) = V_0 \otimes W_0(I - YU^\top)$	$p(H) = \mathcal{N}(H_0, W_0 \otimes W_0)$ $p(Y, S H) = \delta(S - HY)$ $= \lim_{\gamma \rightarrow 0} \mathcal{N}(\vec{S}; \vec{AY}, \gamma^2(I_N \otimes I_M))$ $\mathbb{E}_{ Y}(H) = H_0 + \underbrace{(S - H_0 Y)}_{=: \Delta} \underbrace{(Y^\top W_0 Y)^{-1} Y^\top W_0}_{=: U^\top}$ $+ U\Delta^\top - UY^\top \Delta U^\top$ $\mathbb{V}_{ Y}(H) = W_0(I - YU^\top) \otimes W_0(I - YU^\top)$
Model for A	$p(A) = \mathcal{N}(A_0, V_0 \otimes W_0)$ $p(Y, S A) = \delta(Y - AS)$ $= \lim_{\gamma \rightarrow 0} \mathcal{N}(\vec{Y}; \vec{AS}, \gamma^2(I_N \otimes I_M))$ $\mathbb{E}_{ Y}(A) = A_0 + (Y - A_0 S) \underbrace{(S^\top W_0 S)^{-1} S^\top W_0}_{=: U^\top}$ $\mathbb{V}_{ Y}(A) = V_0 \otimes W_0(I - SU^\top)$	$p(A) = \mathcal{N}(A_0, W_0 \otimes W_0)$ $p(Y, S A) = \delta(Y - AS)$ $= \lim_{\gamma \rightarrow 0} \mathcal{N}(\vec{Y}; \vec{AS}, \gamma^2(I_N \otimes I_M))$ $\mathbb{E}_{ Y}(A) = A_0 + \underbrace{(Y - A_0 S)}_{=: \Delta} \underbrace{(S^\top W_0 S)^{-1} S^\top W_0}_{=: U^\top}$ $+ U\Delta^\top - US^\top \Delta U^\top$ $\mathbb{V}_{ Y}(A) = W_0(I - SU^\top) \otimes W_0(I - SU^\top)$

gives rise to a low-rank mean estimate for A , thus an easily computable estimate for the inverse A^{-1} . In either model class, we can also decide to explicitly model the symmetry of the matrix, or to not encode this information in the probability measure. Modelling symmetry complicates the derivations somewhat, but turns Algorithm 17.2 into a conjugate direction method.

Before making an explicit link between certain classes of classic solvers and particular choices within those four families, we point out some subtle connections and differences between these for model choices.

Although the observation likelihoods in all four cases are Dirac point masses on a projection AS or HY , these point masses arise from different limit processes: If the prior encodes symmetry, i.e. uses the symmetric Kronecker product $\otimes$, then the likelihood has to also involve a symmetric Kronecker product in the covariance – otherwise the posterior does not inherit the symmetric property, and both the posterior mean and covariances do not have the compact forms listed in Table 19.1. Without going into further details, this is the principal reason why it is not straightforward to extend this framework to the noisy setting, i.e. a situation in which matrix-vector multiplications can only be computed with (Gaussian) noise.

Table 19.1: A summary of the four difference classes of probabilistic models under consideration for the construction of linear solvers. Each cell lists the form of the prior over matrix elements, the observation likelihood, and the posterior mean and covariance over the matrix elements thus arising. See text for a discussion.

► 19.8 Consistency between beliefs on the matrix and its inverse

As we have already noted in §19.2, there are structural differences between a Gaussian model over A and one over its inverse H . These differences reflect a larger point that one should be very careful when considering the inverse of Gaussian random variables $x \sim \mathcal{N}(\mu, \sigma^2)$ if the signal-to-noise ratio is small, $\mu/\sigma < 1$ (see Figure 19.9). But if the ratio is large, the inverse is relatively well approximated by a Gaussian itself. What does this mean for our matrix-valued Gaussian beliefs? Are there situations in which a Gaussian prior on the matrix A can be associated with a reciprocal Gaussian belief on H ; and does this relationship persist as observations arrive and the posterior evolves? This section summarises some such results.¹⁹

Definition 19.11 (Posterior correspondence). *Consider two solvers of the form of Algorithm 17.2, one with a belief on A with prior mean A_0 and a covariance parameter W_0^A (with associated posterior mean A_M) and one maintaining a distribution on H with prior mean H_0 and covariance parameter W_0^H (with associated posterior mean H_M). We say their priors induce posterior correspondence if*

$$A_M^{-1} = H_M \quad \text{for } 0 \leq M \leq N. \quad (19.28)$$

And we speak of weak posterior correspondence if we only have

$$A_M^{-1}Y = H_M Y. \quad (19.29)$$

For asymmetric models, it is then possible to first show a general result:

Lemma 19.12. *Assume an asymmetric prior. Let $1 \leq M \leq N$, W_0^A , W_0^H symmetric positive definite, and assume $A_0^{-1} = H_0$. Then prior correspondence (Eq. (19.28)) holds if, and only if,*

$$0 = (AS - A_0S)[(S^\top W_0^A A_0^{-1} AS)^{-1} S^\top W_0^A A_0^{-1} - (S^\top A^\top W_0^H AS)^{-1} S^\top A^\top W_0^H].$$

For example, a specific choice that fits this structure is $A_0 = \alpha_0 I$, $W_0^A = \beta_0 A$ paired with $H_0 = \alpha_0^{-1} I$, $W_0^H = \beta_0/\alpha_0$. A more general, and more practical form, will be developed in §21.

Symmetric models are more restricted. But we can still show weak correspondence:

Theorem 19.13. *Assume an symmetric prior. Let $1 \leq M \leq N$, W_0^A , W_0^H symmetric positive definite, and assume $A_0^{-1} = H_0$. Further*

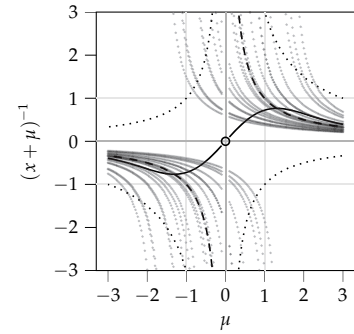


Figure 19.9: The inverse x^{-1} of a scalar Gaussian random variable $x \sim \mathcal{N}(\mu, 1)$ has a complicated distribution if $|\mu| \lesssim 1$ (see also Figure 19.1). For $\mu \neq 0$, the mean (solid line) is given, in a principal value sense, by Dawson's function (Lecomte, 2013)

$$\begin{aligned} \mathbb{E}_{\mathcal{N}(\mu, 1)}(x^{-1}) &= \sqrt{\frac{\pi}{2}} e^{-(x-\mu)^2/2} \operatorname{erfi}\left(\frac{x-\mu}{\sqrt{2}}\right), \end{aligned}$$

where $\operatorname{erfi}(z) = -i \operatorname{erf}(iz)$. However, for $\mu \gg 1$, the distribution $p(x^{-1})$ is relatively well approximated by $\mathcal{N}(x^{-1}; \mu^{-1}, \mu^{-2})$ (dashed line μ^{-1} , dotted lines at $\mu \pm 2\sqrt{\mu^{-2}}$). The plot also shows 20 samples $(x + \mu)^{-1}$. For $\mu = 0$, the mean does not exist.

¹⁹ These statements are all from Wengert and Hennig (2020), where proofs can also be found.

assume that W_0^A, A_0, W_0^H satisfy

$$W_0^A S = Y, \quad \text{and} \\ S^\top (W_0^A A_0^{-1} - A W_0^H) = 0.$$

Then weak posterior correspondence (Eq. (19.29)) holds.

► 19.9 Gaussian Models Consistent with Classic Solvers

By inspecting the above in light of the theorems in §17 onward, we can now establish the following results.

Theorem 19.14 (Probabilistic projection methods). *Any Gaussian generative model on elements of A , $p(A) = \mathcal{N}(A, A_0, \Sigma_A)$ or on elements of the inverse H , $p(H) = \mathcal{N}(H, H_0, \Sigma_H)$, when used in the form of Algorithm 17.2, gives rise to a projection method.*

Proof. By Eq. (19.7), the estimator arising from these priors obeys the consistency requirement for line 12 of Algorithm 17.2 and is thus a projection method by construction. $\square$

Theorem 19.15 (Probabilistic conjugate direction methods). *Any Gaussian generative model with symmetric Kronecker covariance for either A or H , i.e. $p(A) = \mathcal{N}(A, A_0, W_A \otimes W_A)$ or $p(H) = \mathcal{N}(H, H_0, W_H \otimes W_H)$, with symmetric positive definite A_0, W_A or H_0, W_H , when used in Algorithm 17.2, gives rise to a conjugate direction method.*

Proof. The theorem follows immediately from Theorem 18.2, using the consistency of the Gaussian posterior (19.7), and the fact that the posterior estimates A_M^{-1} or H_M are symmetric. $\square$

Theorem 19.16 (Probabilistic conjugate gradients). *Consider a prior $p(H) = \mathcal{N}(H; H_0, W_H \otimes W_H)$. For all parameter choices of (H_0, W_H) with scalar $H_0 = \alpha I$ and $W_H = \beta I + \gamma H$, for $\alpha \in \mathbb{R}$ and $\beta, \gamma \in \mathbb{R}_+$, Algorithm 17.2 is equivalent to the method of conjugate gradients, in the sense that it produces the exact same sequence of estimates x_i . The same is true for the model class $p(A) = \mathcal{N}(A; A_0, W_A \otimes W_A)$ with scalar $A_0 = \alpha I$ and $W_A = \beta I + \gamma A$.²⁰*

²⁰ Yes, W_H includes the true, inaccessible, H , and, similarly, W_A includes A . This oddity is discussed in more depth in §20.

Proof. We will make use of Theorem 18.4. First, by Theorem 19.15, the algorithm constructed here is a conjugate directions method. For Theorem 18.4 to hold, we have to show that the estimator

H_i resulting from these models and parameter choices satisfies Assumption (18.5), i.e. that the gradient r_{i-1} is mapped to $s_i = -H_{i-1}r_{i-1}$ in the span of $\{S, Y, r_{i-1}\}$. We verify that this is true by considering the image of H_i as listed in the right column of Table 19.1. For the model on H , if $S_{:i-1}$ denotes $(s_1, \dots, s_{i-1})$ and analogously for $Y_{:i-1}$, then H_{i-1} maps any vector $v \in \mathbb{R}^N$ to the span of $\{H_0 v, S_{:i-1}, H_0 Y_{:i-1}, W_H Y_{:i-1}\}$. Hence, if H_0 is scalar and $W_H = \beta I + \gamma H$, then (since $HY = S$), r_{i-1} is mapped to the span of $\{r_{i-1}, S_{:i-1}, Y_{:i-1}\}$.

For models on A , H_{i-1} maps r_{i-1} to the span of

$$\{A_0^{-1}r_{i-1}, A_0^{-1}W_A S_{:i-1}, A_0^{-1}Y_{:i-1}, S_{:i-1}\}.$$

For a scalar A_0 and $W_A = \beta I + \gamma A$, using $AS = Y$, this is the span of $\{r_{i-1}, Y_{:i-1}, S_{:i-1}\}$. This concludes the proof. $\square$

Corollary 19.17 (Probabilistic preconditioned conjugate gradient). *Consider a prior $p(A) = \mathcal{N}(A; \alpha K, \beta^2 K \otimes K)$ with a positive definite matrix K that can be decomposed as $K = C^\top C$, and scalars $\alpha, \beta \in \mathbb{R}$. Then Algorithm 17.2 is equivalent to the preconditioned method of conjugate gradients with the preconditioner K . The same is true for the model class $p(H) = \mathcal{N}(H; \tilde{\alpha} K^{-1}, \tilde{\beta}^2 (K \otimes K)^{-1})$.*

Proof. This result follows from Theorem 19.16, by noting that preconditioned CG amounts to running CG on the transformed problem $(C^{-1\top} A C^{-1}) Cx = C^{-1\top} b$. If we define the transformed quantities $\tilde{A} := C^{-1\top} A C^{-1}$, $\tilde{x} = Cx$ and $\tilde{b} = C^{-1\top} b$, then by Theorem 19.16, running conjugate gradient on $\tilde{A}\tilde{x} = \tilde{b}$ is equivalent to running Algorithm 17.2 to infer $\tilde{A}$ from the prior $p(\tilde{A}) = \mathcal{N}(\tilde{A}; \alpha I, \beta^2 (I \otimes I))$. We note that this transformation of A can be written as

$$A = (C \otimes C)^\top \tilde{A}.$$

Thus, by Eq. (3.4), the associated belief over A is a Gaussian of the form²¹

$$\begin{aligned} p(A) &= \mathcal{N}(A; \alpha (C \otimes C)^\top I, \beta^2 (C \otimes C)^\top (I \otimes I) (C \otimes C)) \\ &= \mathcal{N}(A; \alpha K, \beta^2 K \otimes K). \end{aligned}$$

An analogous computation for $H = (C \otimes C)^{-1} \tilde{A}^{-1}$ yields the corresponding prior for H . $\square$

The above results, in particular Theorem 19.16, show that there is an interesting class of probabilistic interpretations for widely used iterative linear solvers. In the sense of these Theorems, we

²¹ The step from the first to the second line can be shown from the definitions of the symmetric (Eq. (19.16)) and standard (Eq. (15.2)) Kronecker products.

may interpret an algorithm like CG as the policy of an agent with an internal probabilistic model for the matrix A , or its inverse H , who explicitly uses this model to iteratively estimate the solution of the linear problem $Ax = b$, and performs probabilistic inference from the observations collected in this process.

It may seem questionable that the exact A , or its inverse H , show up as possible parameter choices in Theorem 19.16. After all, these are the objects of interest in the inference scheme, and thus definitely not available at runtime. For the moment, we just note in passing that the posterior *means* H_M or A_M only contain W_H or W_A in the form $W_H Y$ and $W_A S$, respectively. Since we know $HY = S$ and $AS = Y$, respectively, we can replace $W_H Y = (\beta I + \gamma H)Y = \beta Y + \gamma S$ and, analogously $W_A S = (\beta I + \gamma A)S = \beta S + \gamma Y$. So the means, the point estimators, can be computed for these choices – they only contain aspects of the intractable W that are accessible at runtime (i.e. not A or H).

In contrast, the posterior covariances explicitly contain the matrix W . So if these error estimates are required, this “trick” cannot be used, at least not without further thought. A way to address this issue will be considered in §21. The principal idea will be that of empirical Bayesian inference: If the algorithm has already run for several iterations, and we know what structure it needs to have to be equivalent to CG, what is our best guess for a covariance matrix W that is both consistent with the algorithms’ actions so far and also might preserve the equivalence in future steps? Of course, this will require some assumption of regularity about the matrix A itself.

